

COMPX518-21A

Assignment 3

Total Marks: 100: Weightage 25%

Due: 23rd June, 5 PM, 2021

Password Manager (25 marks)

In this part you are going to design and develop a secure password manager. Your password manager will have the ability to 1) Generate random passwords 2) Securely store the passwords and will consist of at least these modules.

- 1) Password generator module
 - This module will request the length of the password from the user (8-80 characters)
 - The module will then generate random passwords with alphanumeric characters and the special characters (!, @, #, \$, %, ^, &, /, ?, <, >, *, etc. }
- 2) Password storage module
 - The password generated by the generator module will need to be stored securely.
 - The password storage module will request the user for an identifier for the newly generated password (a unique number or the name of the application you will use the password for etc.) and a master password.
 - The module will authenticate the user with the master password and then securely store the new password. The identifier is used to help in password retrieval.
- 3) The password retriever module
 - This module authenticates the user with the master password.
 - Once the user is authenticated the user supplies the identifier of the password it wants to retrieve.
 - The module then retrieves the securely stored password and shows it to the user

You are required to create an architectural diagram that depicts the design of the software. The diagram will show the various entities in the software (modules, submodules etc.) along with the data flow between them and then implement this system. You will also write a report that describes your architecture with clear explanation for

1. Your choice of the random number generator used and the justification for your choice.
2. How you authenticate using the master password with comment on the security of authentication subsystem
3. How you store the passwords generated by the software with comment on the security of the storage subsystem

You will be marked on

1. **(10 marks)** The design of your system
2. Your report
 - a. **(2.5 marks)** Description of the design
 - b. **(2.5 marks)** Choice of PRNG and the justification

- c. **(2.5 marks)** Master password authentication and comment
 - d. **(2.5 marks)** Password storage and comment
- 3. Source Code
 - a. **(1 mark)** Coding style
 - b. **(1 mark)** Logical easy to follow comments
 - c. **(1 mark)** Input Handling
 - d. **(1 mark)** Exception Handling
 - e. **(1 mark)** Overall design

Assignment submission

The assignment submission will have two parts. A document that has the architecture diagram and the report and a zipped file containing the code. Moodle will only accept two files, do not upload individual code files.

Extensions

No extensions will be given unless approved by the Department of Computer Science (<https://www.cs.waikato.ac.nz/student-resources/application-for-an-extension-of-deadline>)
You can submit late. However, late submissions will be deducted **5 marks/ day**.

Plagiarism

Credit all sources you refer to. Students found plagiarising will be reported to the disciplinary committee. You are expected to follow the University's guidelines here: <https://www.waikato.ac.nz/students/academic-integrity/student-information/plagiarism>.
Assignments will be checked against anti-plagiarism checkers.